Esther Mosaku

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Education

University of Puget Sound Tacoma, WA

Bachelor of Science in Computer Science and Economics *Expected May 2028*

Relevant Coursework: Data Structures, Algorithms, Database Systems, Computer Architecture, Linear Algebra, Software Engineering, Statistics, Operating Systems (Spring 2027), AI (Spring 2027)

Awards: Matelich Scholar (Full merit scholarship, 0.67% acceptance rate), GHC'26 Kamala Scholar, Dean's List x3

Experience

Full Stack Software Engineering Intern May 2026 – July 2026

Intuit Credit Karma *Charlotte, NC*

- Built an **AI-powered** pipeline that caught data quality issues before production, replacing a fully manual review process **at scale**.
- Combined rule-based checks, cryptographic change detection, and **LLM**-based semantic analysis in a 5-stage **TypeScript** pipeline processing **700+ data fields**.
- Shipped a severity-based alerting system integrated with **Slack** and a **Next.js** dashboard, catching **300+ P3 and 70+ P2 issues** across the product catalog; orchestrated via **Airflow**, **BigQuery**, and **CircleCI**.

Computer Science Undergraduate Research Assistant May 2025 – August 2025

University of Puget Sound Computer Science Department *Tacoma, WA*

- Built graph-based **RAG** methods that reduced hallucinations in medical **LLMs**, targeting a **30% accuracy gain** (ROUGE, F1) in clinical Q&A.
- Built knowledge graphs from 30+ anonymized medical documents using **Java** and **Stanford CoreNLP**, with a **Python/GPT-4o** evaluation pipeline.
- Presented findings at **IEEE MIT Undergraduate Research Technology Conference (URTC)** and the Murdock College Science Research Conference.

Computer Science Lab Teaching Assistant January 2026 – May 2026

University of Puget Sound *Tacoma, WA*

- Supported weekly lab sessions for 10+ students, debugging code and translating core programming concepts into clear, one-on-one explanations.

Technical Skills

Languages: C, Java, Python, JavaScript, TypeScript, SQL, R, Assembly, HTML/CSS
Frameworks & Libraries: React, Next.js, Scikit-learn, Pandas, Matplotlib, CoreNLP, Firebase
Tools & Systems: Git/GitHub, Linux (Ubuntu), Google BigQuery, Google Cloud Platform, CircleCI (CI/CD), Cursor & Claude Code

Projects

Snake | *C, ncurses, POSIX Sockets* | GitHub

- Built a terminal game in **C** with a custom slab allocator (**3,000-node pool**, intrusive free list) replacing malloc/free, achieving **O(1) alloc/free** with zero fragmentation, validated via **AddressSanitizer**.
- Implemented a persistent leaderboard via binary file I/O and a client-server multiplayer architecture over **TCP sockets** with a custom binary wire protocol.

Safe Inbox (Email Phishing Detector) | *Python, Pandas, Scikit-learn, Matplotlib* | GitHub

- Engineered a supervised **ML** model achieving **95% accuracy** in classifying phishing URLs using feature engineering, vectorization, and hyperparameter tuning.
- Processed **800K+ records**, trained and compared models (**Random Forest**, **XGBoost**, **Logistic Regression**) to optimize detection accuracy.

Leadership & Involvement

Vice President, Girls Who Code College Loop (Sep 2025 - Present) - Co-founded the campus chapter, launching CS trivia nights, web-building workshops, and resume development sessions for a 5-member core team.

President, Math Club (Sep 2025 - May 2026) - Grew membership by 5-10 students through engaging events and cross-club collaborations with the Physics Club.

Organizations & Affiliations

Organizations: ColorStack, Code2040, Rewriting the Code, AI4ALL Ignite Fellow, Volunteer at Habitat for Humanity, Technovation Girls Student Mentor
Additional Honors: EducationUSA Opportunity Funds Scholar (Top 0.175%), Technovation Girls Regional Winner for Africa 2023